CLH-MAX: A Lattice-Based KEM Proposal over the Prime Cyclotomic Ring $\mathbb{Z}_q[X]/\Phi_{257}(X)$

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Abstract

We propose CLH-MAX, a Module-LWE-based key encapsulation mechanism (KEM) in the style of Kyber/ML-KEM in which the ring $\mathbb{Z}_q[X]/(X^{256}+1)$ is replaced by $R_q = \mathbb{Z}_q[X]/\Phi_{257}(X)$, the ring of the prime cyclotomic field $\mathbb{Q}(\zeta_{257})$. For $q = 3329$ the polynomial Φ_{257} is irreducible modulo q, so $R_q \cong \mathbb{F}_{q^{256}}$ is a field. We specify the CPA-secure public-key encryption layer, its Fujisaki–Okamoto transform to an IND-CCA2 KEM (random-oracle model), and three parameter sets of module rank $2, 3, 4$. We give a game-hopping proof that IND-CPA security reduces to the decision Module-LWE problem over R_q, an exact analysis of the decryption failure probability (which corrects the naïve coefficient-variance count for this ring), lattice-estimator outputs for the generic MLWE instances underlying the parameter sets (identical to those of ML-KEM-768/1024 for $k = 3, 4$, as expected, and about 7 bits lower for $k = 2$ when $\eta = 2$ is used), and a discussion of arithmetic cost, where the absence of a negacyclic NTT is the main disadvantage. We make no claim of higher concrete security than ML-KEM; the contribution is a fully specified alternative ring instantiation together with its correctness and reduction analysis, reproducible scripts, and a clear list of open problems.

1 Introduction

ML-KEM (FIPS 203 [9]) is based on Module-LWE over $\mathbb{Z}_{3329}[X]/(X^{256}+1)$, a power-of-two cyclotomic ring chosen because it admits a fast (incomplete) number-theoretic transform (NTT). The choice of ring in structured-lattice cryptography is a design parameter that is not fully explored: other cyclotomic rings have been proposed, e.g. prime-conductor rings in the ring-LWE literature and in NTRU Prime. This note studies one concrete alternative, conductor 257 (the Fermat prime $F_3 = 2^8 + 1$), for which the ring degree is $\varphi(257) = 256$, equal to that of ML-KEM, so module ranks, key sizes and noise parameters can be compared directly.

Contributions. (1) A complete specification of CLH-MAX: PKE, FO-based KEM, three parameter sets (Sections 3–4). (2) A proof that IND-CPA security reduces to two decision-MLWE instances over R_q, with the explicit simulators (Section 5, Appendix B). (3) An exact computation of the decryption failure probability including the ring-specific coefficient count for Φ_{257}; it shows that the highest parameter set does *not* reach 2^{-128} without modification (Section 6, Appendices A,C). (4) A cost analysis of multiplication in R_q (Section 7).

Non-claims. We do not claim that CLH-MAX is more secure than ML-KEM, that known attacks on ML-KEM fail to apply to a field ring, or that worst-case reductions give concrete security at these parameters. Section 5 explains why.

Related work. Ring-LWE over general cyclotomic rings was introduced in [8], including the worst-case reduction for arbitrary cyclotomic fields. Kyber [3] and ML-KEM use module structure over $\mathbb{Z}_q[X]/(X^{256}+1)$; the module variant over $\mathbb{Z}_q[X]/\Phi_{257}$ studied here follows the same template. Subfield and structural attacks are discussed in [1, 4, 5, 6].

2 Preliminaries

Notation. $N = 256$, q prime, $R_q = \mathbb{Z}_q[X]/\Phi_{257}(X)$ with $\Phi_{257} = X^{256} + \cdots + X + 1$. Elements are identified with coefficient vectors in $\mathbb{Z}_q^{256}$. For $k \geq 1$, vectors and matrices over R_q are written with the module rank k as dimension. ψ_η is the centered binomial distribution on $\mathbb{Z}$: $\sum_{i=1}^{\eta}(a_i - b_i)$ with $a_i, b_i \leftarrow \{0,1\}$ uniform; it has variance $\eta/2$. For $\eta = 2$ the variance is 1 (we use $\eta = 2$ throughout, so $\sigma^2 = 1$). χ denotes ψ_2 applied coefficient-wise.

Definition 1 (Decision MLWE)**.** *Let $n, m \geq 1$.* $\mathrm{MLWE}_{n,m,\chi}$ *asks to distinguish $(B, Bs+e)$ from (B, w), where $B \leftarrow U(R_q^{m\times n})$, $s \leftarrow \chi^n$, $e \leftarrow \chi^m$, $w \leftarrow U(R_q^m)$.* $\mathrm{Adv}^{\mathrm{MLWE}}_{n,m,\chi}(\mathcal{D})$ *is* $|\Pr[\mathcal{D}(B, Bs+e) = 1] - \Pr[\mathcal{D}(B, w) = 1]|$.

Definition 2 (PKE, IND-CPA)**.** *A PKE is* $(\mathsf{KeyGen}, \mathsf{Enc}, \mathsf{Dec})$*. In the IND-CPA game the challenger samples (pk, sk) and a bit b; the adversary picks m_0, m_1 of equal length, receives $\mathsf{Enc}(pk, m_b)$ and outputs b'. The advantage is $|\Pr[b' = b] - \frac{1}{2}|$.*

Definition 3 (KEM, IND-CCA2)**.** *A KEM is* $(\mathsf{KeyGen}, \mathsf{Encaps}, \mathsf{Decaps})$*, with* $\mathsf{Encaps}(pk) \to (c, K)$*. In the IND-CCA2 game the adversary receives (c^*, K_b) with K_0 the real key and K_1 uniform, and may query* Decaps *on any $c \neq c^*$; advantage $|\Pr[b' = b] - \frac{1}{2}|$.*

3 The ring

Proposition 4. *Let $q \neq 257$ be prime. Then Φ_{257} is irreducible over $\mathbb{F}_q$ if and only if* $\mathrm{ord}_{257}(q) = 256$*, i.e. q is a primitive root modulo 257. In that case $R_q \cong \mathbb{F}_{q^{256}}$ and $q\mathcal{O}_K$ is a prime ideal of $\mathcal{O}_K = \mathbb{Z}[\zeta_{257}]$.*

Proof. For $\gcd(q, m) = 1$ all irreducible factors of Φ_m over $\mathbb{F}_q$ have degree $\mathrm{ord}_m(q)$; here $\deg \Phi_{257} = \varphi(257) = 256$. The statement about $q\mathcal{O}_K$ is the standard description of inert primes in cyclotomic fields. □

Direct computation gives $\mathrm{ord}_{257}(3329) = \mathrm{ord}_{257}(12289) = 256$ ($3329 \equiv 245$ and $12289 \equiv 210 \pmod{257}$). Among primes $3000 < q < 20000$ there are 922 moduli with this property (e.g. $3001, 3019, 3037, 3041, \ldots$). Not every NTT-friendly prime works: $\mathrm{ord}_{257}(7681) = 128$, so Φ_{257} splits into two factors of degree 128 modulo 7681. We use $q = 3329$, the ML-KEM modulus.

Remark 5 (Comparison with ML-KEM). For $q = 3329$ one has $q \equiv 1 \pmod{256}$ but $q \not\equiv 1 \pmod{512}$ ($3329 \bmod 512 = 257$, $\mathrm{ord}_{512}(3329) = 2$). Hence $X^{256} + 1$ factors modulo 3329 into 128 *quadratic* factors, not 256 linear ones; this is why ML-KEM uses an incomplete, 7-layer NTT. $q = 12289$, which splits $X^{256} + 1$ completely, is not an ML-KEM modulus.

Remark 6 (Subfields). $\mathbb{F}_{q^{256}}$ is not "subfield-free": it contains the subfields $\mathbb{F}_{q^{2^j}}$, $0 \le j \le 7$, and $\mathrm{Gal}(\mathbb{F}_{q^{256}}/\mathbb{F}_q)$ is cyclic of order 256; likewise $\mathbb{Q}(\zeta_{257})$ has a cyclic Galois group of order 256 and a subfield of every degree 2^j. What differs from $X^{256}+1$ is only that R_q does not decompose as a product of smaller rings. Whether this has any security consequence is open.

Multiplication. Let $a, b \in R_q$ with coefficient vectors of length 256 (degree ≤ 255). Then $ab = \sum_{i=0}^{510} c_i X^i$ over $\mathbb{Z}_q$. Since $\Phi_{257} \mid X^{257}-1$, first fold $p_i = \sum_{j+l\equiv i \ (257)} a_j b_l$ for $0 \le i \le 256$; then $ab \bmod \Phi_{257}$ has coefficients $p_i - p_{256}$ for $0 \le i \le 255$ (because $X^{256} \equiv -(X^{255}+\cdots+1)$). Hence a ring multiplication is a length-257 cyclic convolution followed by 256 subtractions.

4 Scheme

Fix $q = 3329$, $\eta = 2$, module rank $k \in \{2,3,4\}$. Let $\mathrm{Enc}_1(m) = \lceil q/2 \rfloor \cdot m$ embed $m \in \{0,1\}^{256}$ coefficient-wise into R_q and $\mathrm{Dec}_1(x)_i = 1$ iff the centered representative of x_i lies in $[\lceil q/4 \rceil, \lfloor 3q/4 \rfloor]$.

PKE.

- KeyGen: $A \leftarrow U(R_q^{k\times k})$; $s, e \leftarrow \chi^k$; $t = As + e$. $pk = (A, t)$, $sk = s$.
- $\mathsf{Enc}(pk, m; r)$: coins determine $r, e_1 \leftarrow \chi^k$, $e_2 \leftarrow \chi$; $u = A^T r + e_1$; $v = t^T r + e_2 + \mathrm{Enc}_1(m)$. Output (u, v).
- $\mathsf{Dec}(sk, (u, v))$: $m' = \mathrm{Dec}_1(v - s^T u)$.

Correctness. $v - s^T u = \mathrm{Enc}_1(m) + E$ with $E = e^T r + e_2 - s^T e_1$; decryption succeeds iff every coefficient of E has centered absolute value $< \lfloor q/4 \rfloor$ (up to rounding of $q/2$).

KEM (FO with implicit rejection). With hash functions H, G modelled as random oracles and following [7]:

- KeyGen: $(pk, s) \leftarrow \mathsf{PKE.KeyGen}$, $z \leftarrow \{0,1\}^{256}$; $sk_{\mathrm{KEM}} = (s, pk, H(pk), z)$.
- $\mathsf{Encaps}(pk)$: $m \leftarrow \{0,1\}^{256}$; $(\bar{K}, r) = G(m \,\|\, H(pk))$; $c = \mathsf{Enc}(pk, m; r)$; output $(c,\ K = H(\bar{K} \,\|\, H(c)))$.
- $\mathsf{Decaps}(sk, c)$: $m' = \mathsf{Dec}(s, c)$; $(\bar{K}', r') = G(m' \| H(pk))$; if $c = \mathsf{Enc}(pk, m'; r')$ (constant-time comparison) return $H(\bar{K}' \| H(c))$, else return $H(z \| H(c))$ (implicit rejection, also in constant time).

By [7], IND-CCA2 security of this KEM in the random-oracle model follows from IND-CPA security of the PKE together with a sufficiently small decryption failure probability δ; the explicit bound of [7] should be consulted for the concrete loss in terms of the number of oracle queries. We make no quantum-random-oracle claim.

Sizes (no compression). With 12-bit packing: $|pk| = 384k + 32$ B, $|c| = 384(k+1)$ B, $|sk_{\mathrm{KEM}}| = 384k + |pk| + 64$ B. Ciphertext compression as in ML-KEM is not specified here and would change

Set	k	$n = 256k$	$\lvert pk\rvert$	$\lvert c\rvert$	$\lvert sk_{\mathrm{KEM}}\rvert$
CLH-MAX-512	2	512	800	1152	1632
CLH-MAX-768	3	768	1184	1536	2400
CLH-MAX-1024	4	1024	1568	1920	3168

Table 1: Parameter sets and sizes in bytes; $q = 3329$, $\eta = 2$, no ciphertext compression.

both δ and ciphertext size.

5 Security

Theorem 7 (IND-CPA). *For every adversary $\mathcal{A}$ against the PKE there are distinguishers $\mathcal{B}_1, \mathcal{B}_2$ with essentially the running time of $\mathcal{A}$ such that*

$$\mathrm{Adv}^{\text{IND-CPA}}(\mathcal{A}) \le \mathrm{Adv}^{\text{MLWE}}_{k,k,\chi}(\mathcal{B}_1) + \mathrm{Adv}^{\text{MLWE}}_{k+1,k,\chi}(\mathcal{B}_2).$$

Proof. *Game* G_0 is the real IND-CPA game. *Game* G_1 is G_0 with t replaced by a uniform $t^* \in R_q^k$. $\mathcal{B}_1$, given an MLWE challenge (A, t^*) with k samples and secret s, runs $\mathcal{A}$ on $pk = (A, t^*)$, plays the challenger honestly for the rest and outputs 1 iff $\mathcal{A}$ wins. If $t^* = As + e$ this is exactly G_0, otherwise G_1, so $|\Pr[G_0] - \Pr[G_1]| = \mathrm{Adv}^{\text{MLWE}}_{k,k,\chi}(\mathcal{B}_1)$. *Game* G_2 replaces (u, v) by uniform elements of $R_q^k \times R_q$. In G_1 the matrix $B = \binom{A^T}{t^T} \in R_q^{(k+1)\times k}$ is uniform and

$$\begin{pmatrix} u \\ v - \mathrm{Enc}_1(m_b) \end{pmatrix} = B\, r + \begin{pmatrix} e_1 \\ e_2 \end{pmatrix}$$

is an MLWE sample with secret r and $k+1$ rows. $\mathcal{B}_2$, given (B, w), parses B into A^T and t^T, runs $\mathcal{A}$ on $pk = (A, t)$, and after the challenge messages are chosen returns $\mathcal{A}$ the ciphertext $(w_{1..k},\ w_{k+1} + \mathrm{Enc}_1(m_b))$. Real samples give G_1, uniform ones G_2. In G_2 the ciphertext is independent of b so $\Pr[G_2] = \frac{1}{2}$. Combining, $\Pr[G_0] - \frac{1}{2} \le \varepsilon_1 + \varepsilon_2$. □

Remark 8 (Why no "axiom"). The theorem is a relative statement. It does *not* give $\mathrm{Adv}^{\text{IND-CPA}} = 0$: an unbounded adversary recovers s from (A, t) (the support of (s, e) is far smaller than q^{256k}), so no sound assumption can set the MLWE advantage of *all* distinguishers to zero. Security is computational and rests on the hardness of MLWE for efficient adversaries. Any mechanised proof should formalise the inequality of Theorem 7, not an unconditional-hardness axiom.

Worst-case reductions. Lyubashevsky–Peikert–Regev [8] give a quantum reduction from approximate worst-case SVP on ideal lattices to RLWE for *any* cyclotomic field, hence including $\mathbb{Q}(\zeta_{257})$, but for discrete Gaussian errors in the canonical embedding with width $\alpha q \ge \omega(\sqrt{\log n})$ and with moduli and approximation factors that are polynomial in n with non-trivial exponents. The parameters used here ($q = 3329$, centered binomial $\eta = 2$, $n = 256$) lie outside that regime, exactly as for ML-KEM, so the reductions provide asymptotic motivation only and no concrete bits of security. Moreover, for prime-power cyclotomic fields (which includes conductor 257) there are quantum algorithms for ideal-SVP with approximation factors $\exp(\tilde{O}(\sqrt{n}))$ in certain settings [5, 6]; these do not break RLWE at the parameters considered, but they caution against reading worst-case statements as guarantees.

Concrete security. Attacks on (M)LWE (primal uSVP, BDD, dual, hybrid) as implemented in the lattice-estimator [2] see only the triple (n, q, χ) with $n = kN$; they do not depend on the ring structure. We ran `LWE.estimate` (default settings, October 2026) with $m = n$, $q = 3329$ on the key-recovery instance of each CLH-MAX parameter set and, with the *same call*, on the corresponding ML-KEM instance. Table 2 reports the output unchanged. The outcome is what the structure of the estimator predicts: for $k = 3, 4$ the instances coincide with ML-KEM-768/1024 and so do the estimates; for $k = 2$ the choice $\eta = 2$ makes CLH-MAX-512 about 7 bits *weaker* than ML-KEM-512 (which uses ψ_3 for the key). Hence the ring $\mathbb{Z}_q[X]/\Phi_{257}$ neither helps nor hurts *in these generic estimates*; nothing in them supports a security advantage over ML-KEM. Limitations: only the key-recovery instance was estimated (the message-recovery instance with secret r should also be run), the estimates are for generic MLWE, and we assign no NIST category. Known structural

Instance $(n;\ \text{secret}, \text{error})$	BKW	uSVP	BDD	Dual	DH	β_{uSVP}	β_{DH}
$(512; \psi_3, \psi_3)$ [ML-KEM-512]	178.8	143.8	140.2	149.9	139.7	406	387
$(512; \psi_2, \psi_2)$ [CLH-MAX-512]	167.2	137.1	133.7	142.6	132.8	382	363
$(768; \psi_2, \psi_2)$ [both]	238.3	204.9	201.0	214.3	196.4	624	589
$(1024; \psi_2, \psi_2)$ [both]	315.0	275.1	270.7	288.5	262.3	874	823

Table 2: lattice-estimator output, $\log_2$(operations), $q = 3329$, $m = n$, one call and one cost model for all rows. DH = dual hybrid, the cheapest attack in every row. Instances for $n = 768, 1024$ are literally identical for CLH-MAX and ML-KEM, hence identical outputs.

attacks (subfield attacks on overstretched NTRU [1, 4]) target large-modulus NTRU-type instances and are not known to apply to MLWE at $q = 3329$ in either ring; Remark 6 explains why the field structure of R_q is not obviously free of exploitable structure.

6 Correctness: decryption failure probability

Lemma 9. *For $a, b \in R_q$ with independent coefficients, every coefficient of $ab \in R_q$ is a signed sum of* 510 *products $a_j b_l$ of distinct index pairs, except coefficient* 255 *which is a signed sum of* 511*.*

The proof is a direct count of index pairs (Appendix A). This is roughly twice the 256 terms of a negacyclic product, so the variance of a product coefficient is about $2\sigma^4 \cdot 256$ rather than $256\sigma^4$.

For module rank k, the coefficient E_i of $e^T r - s^T e_1 + e_2$ is therefore a sum of $2k \cdot 511$ products of two independent ψ_2 samples (a conservative count) plus one ψ_2 sample, with variance $\sigma_E^2 = 1022k + 1$. Following the usual heuristic of treating the summands as independent, we compute the *exact* distribution by convolution and bound $\delta \le 256 \cdot \Pr[|E_i| \ge \lfloor q/4 \rfloor]$ with $\lfloor q/4 \rfloor = 832$. The script is in Appendix C. Sets with $k = 2, 3$ satisfy $\delta < 2^{-128}$; **the set with $k = 4$ does not** ($\delta \approx 2^{-117}$).

Set	k	σ_E^2	$\log_2 \Pr[\lvert E_i \rvert \ge 832]$	$\log_2 \delta$
CLH-MAX-512	2	2045	-239.0	-231
CLH-MAX-768	3	3067	-163.9	-156
CLH-MAX-1024	4	4089	-124.7	-117

Table 3: Decryption failure probability, no ciphertext compression, $q = 3329$, $\eta = 2$.

Using it inside FO would require changing parameters (e.g. a larger modulus with adapted noise, or an error-correcting encoding of m), which would also change the security analysis; we leave this open. The Gaussian tail approximation $\exp(-832^2/2\sigma_E^2)$ gives $\approx 2^{-245}$ per coefficient for $k = 2$, in line with the exact value.

Trade-off between security and correctness at $k = 2$. Raising the key noise of CLH-MAX-512 to ML-KEM-512's $\eta_1 = 3$ (with $r \leftarrow \psi_3$, $e_1, e_2 \leftarrow \psi_2$, so that the key-recovery instance equals the ML-KEM-512 row of Table 2) closes the 7-bit gap but raises σ_E^2 to about 3834; the exact computation gives $\delta \approx 2^{-123}$, which does *not* meet 2^{-128} (ML-KEM-512 achieves its failure rate with only 256 terms per product coefficient, plus ciphertext compression effects not modelled here). Thus for conductor 257 the near doubling of terms per coefficient (Lemma 9) forces a choice at $k = 2$: $\eta = 2$ with $\delta \approx 2^{-231}$ and about $2^{132.8}$ (dual hybrid), or $\eta_1 = 3$ with $\delta \approx 2^{-123}$ and about $2^{139.7}$. We specify the former and regard the latter as an open design question.

7 Arithmetic

By Section 3, multiplication in R_q is a length-257 cyclic convolution followed by a correction. Schoolbook multiplication costs $256^2 = 65{,}536$ coefficient products per ring multiplication; KeyGen needs k^2 ring products for As and Enc needs $k^2 + k + k$ more. A fast NTT route requires a transform of length ≥ 511 covering the linear convolution, i.e. $512 \mid q - 1$. $q = 12289$ satisfies this ($12288 = 3 \cdot 2^{12}$) and $\mathrm{ord}_{257}(12289) = 256$, but $q = 3329$ does not ($3328 = 2^8 \cdot 13$). With $q = 3329$ one must use Karatsuba/Toom–Cook, Bluestein/Rader-type or lifted-modulus methods. No implementation or benchmark is presented and we make no performance claim. Changing to $q = 12289$ would require re-deriving the noise parameters, the security estimates and δ.

8 Comparison with ML-KEM

	CLH-MAX (this work)	ML-KEM (FIPS 203)
Ring	$\mathbb{Z}_{3329}[X]/\Phi_{257}$, field $\mathbb{F}_{q^{256}}$	$\mathbb{Z}_{3329}[X]/(X^{256}+1)$, 128 quadratic factors
Fast multiplication	no negacyclic NTT at $q = 3329$	incomplete 7-layer NTT
Noise	ψ_2 throughout	ψ_3 in the key for ML-KEM-512
Generic MLWE estimate	Table 2: identical to ML-KEM for $k = 3, 4$; ≈ 7 bits lower at $k = 2$	same estimator call
Worst-case motivation	LPR13, outside concrete regime	same
Standardisation / analysis	none (research proposal)	NIST standard, extensive analysis
δ	$< 2^{-128}$ for $k = 2, 3$; $\approx 2^{-117}$ for $k = 4$	see FIPS 203

Table 4: Qualitative comparison; no security ranking is claimed.

9 Artifact availability

The source files of this version (LaTeX source, the Lean 4 development, the scripts of Appendix C and the raw estimator output) are archived on Zenodo:

- version 2.0.0 (this paper): `https://doi.org/10.5281/zenodo.23428067`
- all versions (always resolves to the latest): `https://doi.org/10.5281/zenodo.22883946`

Earlier Zenodo versions (v1 to 1.6) contain the previous version of this work, whose claims are withdrawn; see the file `ERRATA.md` in the repository.

The file `ClhMaxSecurity.lean` compiles against Mathlib v4.35.0 and formalises the game-hopping inequality underlying Theorem 7 for one ring component ($k = 1$): the left-or-right IND-CPA advantage is bounded by the sum of the advantages of four explicit distinguishers against (Ring-)LWE with one and two samples. It uses no hardness axiom and no `sorry`; `#print axioms` reports only `propext`, `Classical.choice` and `Quot.sound`. It does not formalise the hardness of (M)LWE, running times, or the module case $k > 1$.

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10 Open problems

(i) Choose $k = 2$ and $k = 4$ parameters that meet both $\delta < 2^{-128}$ and ML-KEM-level security (the extra terms per product coefficient in R_q cost correctness; ciphertext compression, an error-correcting message encoding or a different modulus are candidates). (ii) Estimate the message-recovery instance as well as key recovery. (iii) Efficient implementation at $q = 3329$. (iv) Study whether the field structure and the order-256 Galois action of R_q give any algebraic leverage against MLWE. (v) A machine-checked proof of Theorem 7.

A Proof of Lemma 9

Coefficient i of $ab \bmod \Phi_{257}$ is $p_i - p_{256}$ with $p_i = \sum_{j+l\equiv i\ (257)} a_j b_l$, $0 \le j, l \le 255$. For $0 \le i \le 255$: pairs with $j + l = i$ number $i + 1$, and pairs with $j + l = i + 257$ number $255 - (i + 2) + 1 = 254 - i$ (for $i \le 254$) and 0 for $i = 255$. So p_i has 255 terms for $i \le 254$ and 256 terms for $i = 255$. For p_{256}: $j + l = 256$ gives 255 pairs ($j = 1, \ldots, 255$); $j + l = 513 > 510$ is impossible, so p_{256} has 255 terms. The index-pair sets of p_i and p_{256} are disjoint, hence the coefficient is a signed sum of $255 + 255 = 510$ products, or $256 + 255 = 511$ for $i = 255$. □

B Simulators for Theorem 7

```
B1(A, t*):                          # MLWE challenge: k samples, secret s
  (m0, m1) <- Adv.choose(pk=(A, t*)); b <- {0,1}
  r, e1 <- chi^k; e2 <- chi
  u = A^T r + e1; v = t*^T r + e2 + Enc1(m_b)
  b' <- Adv.guess(pk, (u, v)); return [b' == b]

B2(B, w):                           # MLWE challenge: k+1 samples, secret r
  parse B = [A^T ; t^T]             # A in R^{k x k}, t in R^k
  (m0, m1) <- Adv.choose(pk=(A, t)); b <- {0,1}
  u = w[1..k]; v = w[k+1] + Enc1(m_b)
  b' <- Adv.guess(pk, (u, v)); return [b' == b]
```

Real MLWE samples reproduce the corresponding game exactly; uniform samples reproduce the next game.

C Reproducibility scripts

Decryption failure probability (exact convolution under the independence heuristic; $e, s, r \leftarrow \psi_{\eta_1}$, $e_1, e_2 \leftarrow \psi_{\eta_2}$).

```
import numpy as np
from math import log2, comb
def cbd(eta):
    return np.array([comb(2*eta, i) for i in range(2*eta+1)], float)/4**eta, eta
def prod_pmf(x, y):
    p1, o1 = cbd(x); p2, o2 = cbd(y); d = {}
    for i, a in enumerate(range(-o1, o1+1)):
```

```
        for j, b in enumerate(range(-o2, o2+1)):
            d[a*b] = d.get(a*b, 0) + p1[i]*p2[j]
    off = max(abs(v) for v in d); arr = np.zeros(2*off+1)
    for v, p in d.items(): arr[v+off] += p
    return arr, off
def conv_pow(p, off, m):
    res, roff, base, boff = np.array([1.0]), 0, p.copy(), off
    while m:
        if m & 1: res = np.convolve(res, base); roff += boff
        m >>= 1
        if m: base = np.convolve(base, base); boff *= 2
    return res, roff
def delta(k, eta1, eta2, thr=832):        # thr = floor(q/4), q = 3329
    pa, oa = prod_pmf(eta1, eta1); pb, ob = prod_pmf(eta1, eta2)
    d1, o1 = conv_pow(pa, oa, k*511); d2, o2 = conv_pow(pb, ob, k*511)
    d, o = np.convolve(d1, d2), o1 + o2
    p2, oo = cbd(eta2); d, o = np.convolve(d, p2), o + oo
    idx = np.arange(len(d)) - o
    t = d[np.abs(idx) >= thr].sum()
    return log2(t), log2(t) + 8           # per coefficient, union over 256
for k, e1, e2 in [(2,2,2), (3,2,2), (4,2,2), (2,3,2)]:
    print(k, e1, e2, delta(k, e1, e2))
```

lattice-estimator runs (same call for all rows; record the estimator commit).

```
from estimator import *
CBD = ND.CenteredBinomial
sets = [("CLH-MAX-512", 512, 2, 2), ("CLH-MAX-768", 768, 2, 2),
        ("CLH-MAX-1024", 1024, 2, 2), ("ML-KEM-512", 512, 3, 3),
        ("ML-KEM-768", 768, 2, 2), ("ML-KEM-1024", 1024, 2, 2)]
for name, n, es, ee in sets:
    p = LWE.Parameters(n=n, q=3329, Xs=CBD(es), Xe=CBD(ee), m=n, tag=name)
    print(name); print(LWE.estimate(p))
```

References

[1] M. Albrecht, S. Bai, L. Ducas. A subfield lattice attack on overstretched NTRU assumptions. CRYPTO 2016.

[2] M. Albrecht, R. Player, S. Scott. On the concrete hardness of learning with errors. J. Math. Cryptology 9(3), 2015.

[3] J. Bos, L. Ducas, E. Kiltz, T. Lepoint, V. Lyubashevsky, J. Schanck, P. Schwabe, G. Seiler, D. Stehlé. CRYSTALS-Kyber: a CCA-secure module-lattice-based KEM. EuroS&P 2018.

[4] J. H. Cheon, J. Jeong, C. Lee. An algorithm for NTRU problems and cryptanalysis of the GGH multilinear map without a low-level encoding of zero. LMS J. Comput. Math. 19(A), 2016.

[5] R. Cramer, L. Ducas, C. Peikert, O. Regev. Recovering short generators of principal ideals in cyclotomic rings. EUROCRYPT 2016.

[6] R. Cramer, L. Ducas, B. Wesolowski. Short Stickelberger class relations and application to Ideal-SVP. EUROCRYPT 2017.

[7] D. Hofheinz, K. Hövelmanns, E. Kiltz. A modular analysis of the Fujisaki–Okamoto transformation. TCC 2017.

[8] V. Lyubashevsky, C. Peikert, O. Regev. On ideal lattices and learning with errors over rings. J. ACM 60(6), 2013.

[9] NIST. FIPS 203: Module-Lattice-Based Key-Encapsulation Mechanism Standard. 2024.